

ALLAN UMAÑA ORTIZ

CIVIL & HYDRAULIC ENGINEER | WATER SYSTEMS DIRECTOR | DIGITAL TRANSFORMATION

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EXECUTIVE PROFILE

Civil and hydraulic engineer with 13+ years of experience spanning infrastructure design, water utility operations, systems optimization and technical leadership. Currently directs metropolitan water-system improvement initiatives at Costa Rica's national water utility (AyA), combining hydraulic engineering with software development, business intelligence, geospatial analysis and operational technology. Experienced in leading multidisciplinary teams, managing multimillion-dollar budgets, translating complex data into actionable decisions and designing practical digital solutions for public-service infrastructure.

SELECTED IMPACT

- Lead improvement and monitoring initiatives serving a metropolitan water-supply network supporting more than 1.4 million people.
- Manage annual budgets exceeding US\$2 million; oversee investment initiatives including a CRC 510 million flow-metering expansion project (BPIP 002926).
- Led sectorization and efficiency initiatives in Alajuelita, Hatillo and Coronado; reported a 30% average reduction in minimum night flow and recovery of approximately 700 m³/day.
- Developed BI and geospatial monitoring solutions integrating operational, cadastral and commercial information; reported a 15% reduction in response time to supply incidents.
- Led a water-capacity assessment of 30 metropolitan systems and supported commissioning of the Zamora source (+250 L/s), contributing to supply stabilization and capacity planning.

LEADERSHIP & TECHNICAL STRENGTHS

Water systems optimization • Hydraulic analysis and water balances • Non-revenue water • Program and budget management • GIS and geospatial analytics • Digital product development • Data integration and automation • SCADA / telemetry • Performance dashboards • Regulatory and audit reporting • Cross-functional coordination

PROFESSIONAL EXPERIENCE

Director of Systems Improvement / Metropolitan Water Systems | AyA, Costa Rica | Oct 2023–Present

- Direct multidisciplinary work on flow and pressure monitoring, hydraulic capacity, water balances, distribution-system efficiency and infrastructure-development assessments across the Greater Metropolitan Area.
- Coordinate investment prioritization, operational assessments, executive reporting and technical responses to regulators, public institutions and oversight bodies.
- Advance digital decision-support capabilities through integrated Power BI dashboards, ArcGIS-based viewers, geospatial datasets and data-driven monitoring workflows.
- Design and prototype tailored web tools for water-capacity analysis, infrastructure needs tracking and flow-meter inspection, combining Python, Streamlit, SQL and modern web interfaces.
- Develop data-management workflows linking enterprise geodatabases, SharePoint/Microsoft Lists and operational review processes; explore secure SCADA-data access and telemetry integration.

Expert Executive, Water Systems | AyA, Costa Rica | Dec 2022–Oct 2023

- Prepared project pre-feasibility studies, supported technical oversight of wells and water transmission works, and formulated infrastructure investments.

- Built analytical models, dashboards and geographic data structures to improve system planning and management decisions.

Specialist / Advanced Executive, Water Systems | *AyA, Costa Rica* | *Oct 2016–Dec 2022*

- Produced hydraulic and civil engineering designs for potable-water projects including Cañas–Bebedero, Sardinal–El Coco–Ocotal and Nimboyores.
- Supported system reliability, climate resilience, arsenic-control initiatives and technical problem-solving for regional water infrastructure.

Independent Hydrology & Hydraulics Consultant | *Costa Rica* | *Oct 2015–Oct 2016*

- Delivered stormwater, water-recovery, treatment and hydraulic solutions for industrial and residential clients, including projects for Gerber–Nestlé (PREVASSA) and PriceSmart.

Junior Civil Engineer | *SANTOS CMI* | *Jul 2013–Jun 2015*

- Contributed to civil infrastructure design involving roads, drainage, pavements and sanitary works, including coordination on projects in Brazil, Chile and Ecuador.

Professor of Hydraulics | *Universidad Latina de Costa Rica* | *Sep 2018–Present*

- Teach Fluid Mechanics and Sanitary Engineering; advise academic projects related to sediments, sewer networks and watershed modeling.

RECENT DIGITAL ENGINEERING INITIATIVES

- Water Capacity & Needs Management Platform — Development of GIS-enabled views and structured workflows for project tracking, water-capacity analysis, investment needs and technical reporting.
- Flow-Meter Inspection & Monitoring Application — Development of a geospatial inspection interface integrating equipment data, interactive maps, review records, operational dashboards and reporting workflows.
- Public Water-Capacity GIS Viewer — Development and enhancement of a Streamlit-based public mapping interface with spatial filters, geodata layers and user-facing analytical tools.
- Enterprise Data Integration — Python/SQL and Microsoft 365 workflows for extracting geodatabase attributes, georeferencing equipment and connecting operational review information to analysis platforms.

TECHNOLOGY STACK

Data & programming: Python, SQL, SQL Server, PostgreSQL, Power BI, data modeling, HTML/CSS/JavaScript, Streamlit, Plotly, Supabase, GitHub.

GIS & automation: ArcGIS Pro / Enterprise, Experience Builder, Survey123, SharePoint Lists, Power Automate, Power Apps, geodatabases.

Engineering & OT: WaterGEMS, HEC-RAS, AutoCAD, Civil 3D, SCADA / telemetry data workflows.

EDUCATION

Professional Master's in Hydraulic Engineering — University of Costa Rica | Coursework completed (63/63 credits); thesis in progress | GPA 9.24/10

Licentiate Degree in Civil Engineering — Universidad Latina de Costa Rica | 2013

LANGUAGES

Spanish — Native | English — Advanced (C1) | Italian — Intermediate / Upper-intermediate (B1–B2)